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EX:1 PLOT A TIME SERIES DATA

AIM

To develop a Python program to plot and visualize time series data such as population, commodity market price, or temperature using Pandas and Matplotlib.

ALGORITHM

1. Start the program.

2. Import the required libraries, Pandas and Matplotlib.

3. Create the time-series data using Year or Month and the corresponding values.

4. Store the data in a Pandas DataFrame.

5. Select the time column and value column.

6. Plot the data using a line graph.

7. Add a suitable title and labels for the X-axis and Y-axis.

8. Display the graph.

9. Observe the graph and identify the trend.

10. Stop the program.

1. Plot annual population for 2015–2025.

PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt

year = [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025]
population = [90, 93, 96, 98, 100, 103, 106, 110, 114, 118, 123]

data = pd.DataFrame({
    "Year": year,
    "Population": population
})

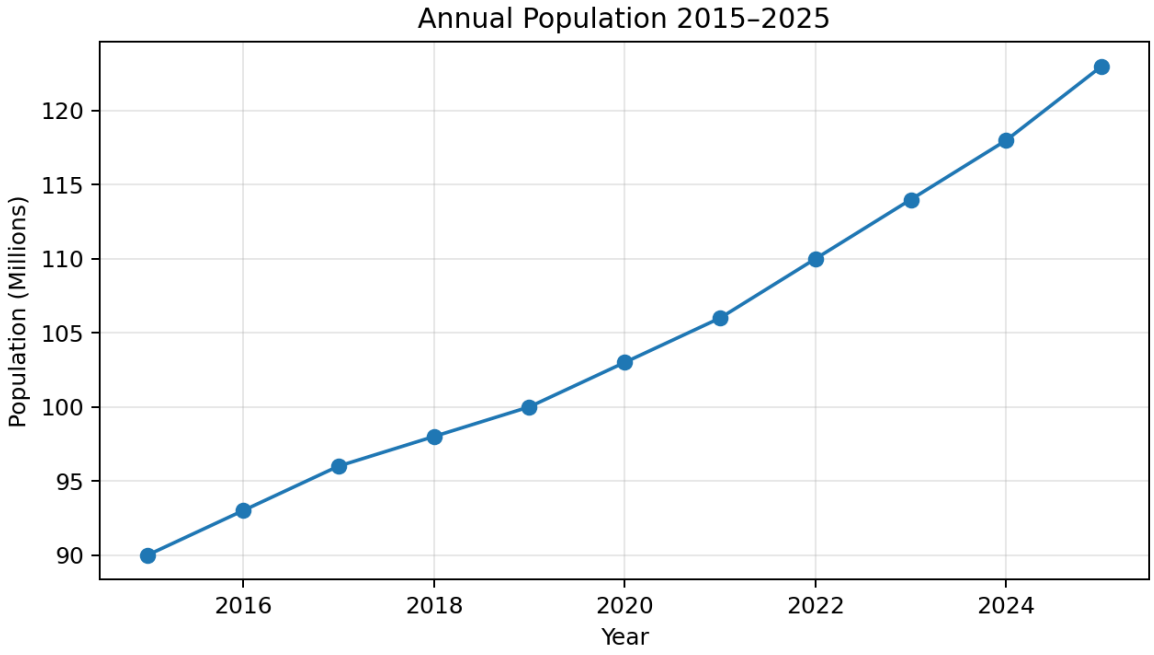
print(data)

plt.plot(data["Year"], data["Population"])
plt.title("Annual Population 2015-2025")
plt.xlabel("Year")
plt.ylabel("Population (Millions)")
plt.show()
```

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OUTPUT – GRAPH

	Year	Population
0	2015	90
1	2016	93
2	2017	96
3	2018	98
4	2019	100
5	2020	103
6	2021	106
7	2022	110
8	2023	114
9	2024	118
10	2025	123



OBSERVATION

The time series shows an increasing trend.

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2. Plot monthly temperature for 12 months.

PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt

month = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
         "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

temperature = [25, 27, 30, 33, 35, 34, 32, 31, 30, 29, 27, 25]

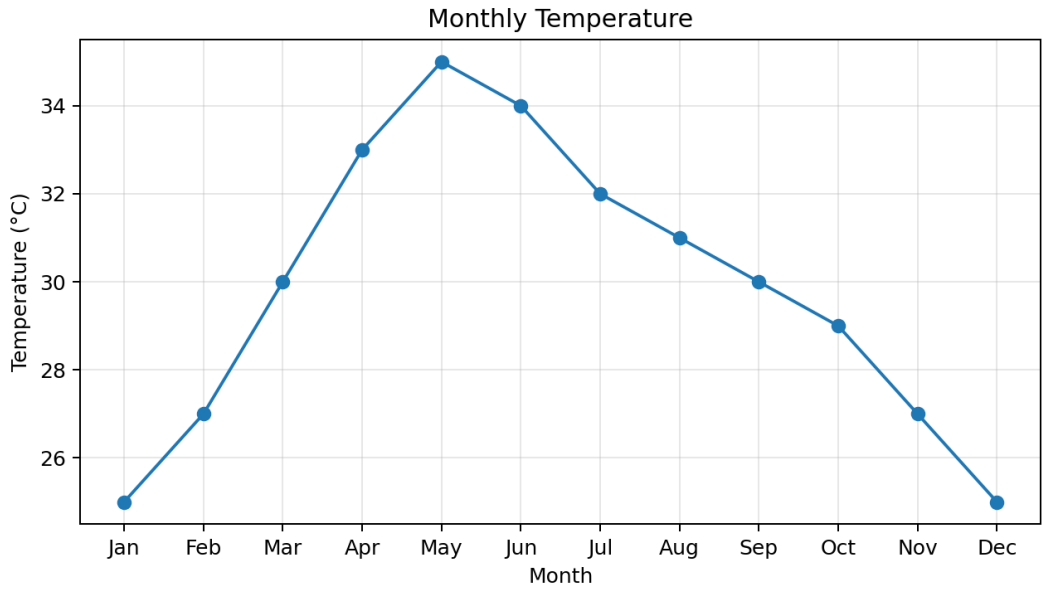
data = pd.DataFrame({
    "Month": month,
    "Temperature": temperature
})

print(data)

plt.plot(data["Month"], data["Temperature"])
plt.title("Monthly Temperature")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.show()
```

OUTPUT – GRAPH

```
    Month  Temperature
0     Jan           25
1     Feb           27
2     Mar           30
3     Apr           33
4     May           35
5     Jun           34
6     Jul           32
7     Aug           31
8     Sep           30
9     Oct           29
10    Nov           27
11    Dec           25
```



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OBSERVATION

The time series shows a seasonal pattern.

3. Plot monthly commodity price for 12 months.

PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt

month = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
         "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

price = [4200, 4350, 4280, 4500, 4650, 4800,
         4700, 4900, 5000, 5150, 5250, 5400]

data = pd.DataFrame({
    "Month": month,
    "Price": price
})

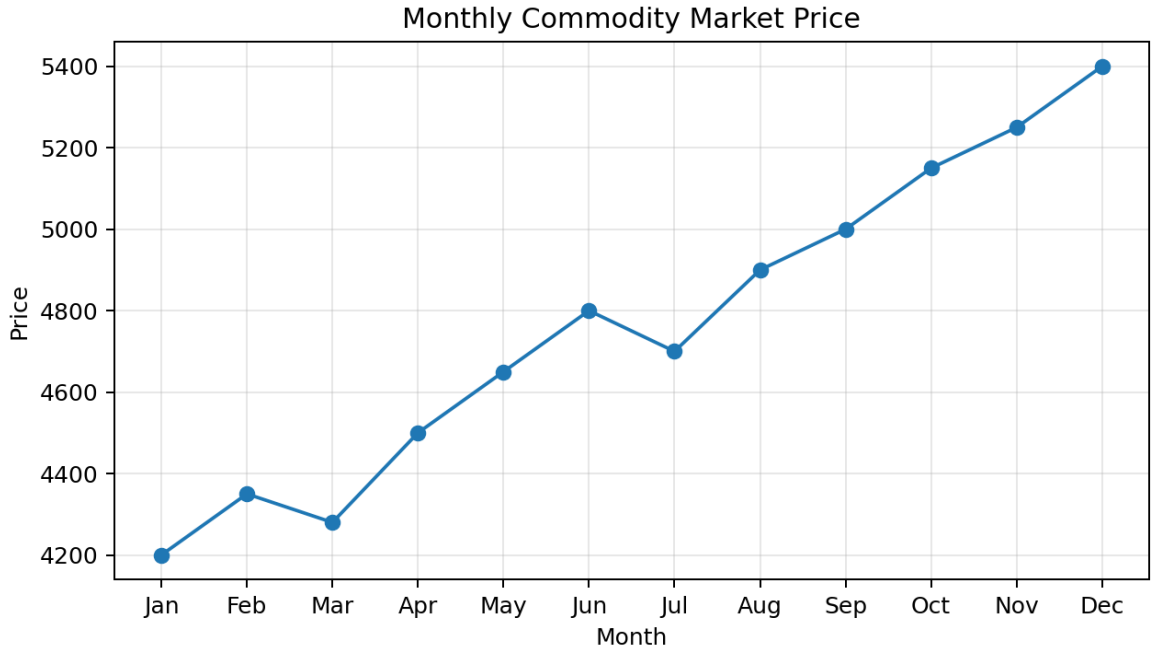
print(data)

plt.plot(data["Month"], data["Price"])
plt.title("Monthly Commodity Market Price")
plt.xlabel("Month")
plt.ylabel("Price")
plt.show()
```

OUTPUT – GRAPH

```
    Month  Price
0     Jan   4200
1     Feb   4350
2     Mar   4280
3     Apr   4500
4     May   4650
5     Jun   4800
6     Jul   4700
7     Aug   4900
8     Sep   5000
9     Oct   5150
10    Nov   5250
11    Dec   5400
```

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OBSERVATION

The time series shows an increasing trend.

RESULT

Thus, the given time-series data were successfully plotted using Python, Pandas, and Matplotlib, and the trends were identified.